Personal Expense Tracker — Project Documentation

Course: React.js Project Assignment
Project option: Option 2 — Personal Expense Tracker
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1. Project overview

Personal Expense Tracker is a single-page React application for logging **income** and **expenses**, viewing a **running balance**, and reviewing spending habits. Transactions are stored in the browser using **localStorage**, so data remains after a refresh. The sidebar shows a **monthly summary** and a **spending-by-category** chart; an optional **monthly budget** displays a warning when spending exceeds the limit.

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2. Features implemented

Feature	Description
Add transactions	Amount, category, description, date; income or expense
Running balance	Current balance = total income − total expenses
Transaction list	All entries with delete per row
Filter & sort	Filter by category; sort by date or amount
Monthly summary	Income, expenses, and net for the selected month
Category chart	Bar chart of expense totals by category (stretch)
Budget alert	Optional monthly budget with over-limit warning (stretch)
Persistence	Transactions and budget saved in localStorage
Responsive UI	Desktop two-column layout; stacked layout on mobile

Stretch goals implemented: category bar chart, monthly summary, budget limit warning.

Not implemented: edit existing transaction in place (delete and re-add instead).

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3. Technologies used

- **React 19** (functional components only)
- **Vite** (dev server and production build)

- **JavaScript (ES modules)**
- **CSS** (custom properties, responsive grid)
- **localStorage** for client-side persistence
- **Oxlint** for linting

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4. Project structure

```
expense-tracker/
├── index.html
├── package.json
├── README.md
├── DOCUMENTATION.md
├── DOCUMENTATION.pdf
├── screenshots/
├── scripts/
│   └── build-docs.mjs
├── src/
│   ├── main.jsx
│   ├── App.jsx
│   ├── App.css
│   ├── index.css
│   ├── constants/
│   │   └── categories.js
│   ├── hooks/
│   │   └── useTransactionStorage.js
│   ├── utils/
│   │   └── transactionHelpers.js
│   └── components/
│       ├── BalanceSummary.jsx
│       ├── TransactionForm.jsx
│       ├── TransactionFilters.jsx
│       ├── TransactionList.jsx
│       ├── TransactionItem.jsx
│       ├── MonthlySummary.jsx
│       └── CategoryChart.jsx
```

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5. How the application works

5.1 Startup (`main.jsx`)

Renders the root `App` component in Strict Mode and loads global styles.

5.2 State and persistence (`App.jsx` + `useTransactionStorage.js`)

- **useState:** transactions, budget limit, filter category, sort order, selected month
- **useEffect:** writes transactions and budget to localStorage whenever they change
- Initial state is read from localStorage on first load

5.3 Helpers (`transactionHelpers.js`)

- **calculateBalance** / **calculateTotals** — income, expenses, and net balance
- **filterAndSortTransactions** — category filter and date/amount sorting
- **spendingByCategory** — totals for the category chart

5.4 Components

Component	Role
BalanceSummary	Shows balance, income, expenses, and budget alert
TransactionForm	Controlled form; toggles income vs expense categories
TransactionFilters	Category, sort, month picker, optional budget
TransactionList	Maps transactions to list items or empty state
TransactionItem	Single row with meta data and delete button
MonthlySummary	Selected month income, expenses, net
CategoryChart	Horizontal bars for expense categories

5.5 Data flow

User submits form → `App.handleAdd` → updates transactions → `useEffect` saves to localStorage → summary, list, and chart re-render from derived values.

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6. Assignment requirements (checklist)

Requirement	Status
Functional React components	Done

4–5+ reusable components	Done (7 components)
Props (parent → child)	Done
useState	Done
useEffect (localStorage sync)	Done
List rendering with keys	Done
Controlled form	Done
Conditional rendering (empty, alerts)	Done
Filter/sort by category or date	Done
localStorage persistence	Done
Responsive layout	Done
README + documentation with screenshots	Done

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7. Setup and run

1. Install dependencies: `npm install`
2. Start development server: `npm run dev`
3. Open the URL shown in the terminal (usually `http://localhost:5173`)
4. Production build: `npm run build`
5. Preview build: `npm run preview`
6. Regenerate PDF docs and screenshots: `npm run docs`

No API keys or `.env` file are required.

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8. Screenshots

Dashboard with sample transactions

REACT COURSE PROJECT

Personal Expense Tracker

Log income and expenses, track your balance, and review spending by category.

Current balance
$2,074.10
Income minus expenses

Total income
$3,450.00

Total expenses
$1,375.90

Add transaction

Expense | Income

Amount
0.00

Category
Food

Description
What was this for?

Date
09/28/2026

Add expense

Filters & budget

Category
All categories

Sort by
Date (newest first)

Month (summary)
September 2026

Monthly budget (optional)
1400

Transactions (6)

Logo design gig +$250.00 Delete
Freelance Sep 15, 2026 Income

Streaming subscription −$38.50 Delete
Entertainment Sep 12, 2026 Expense

Rent share −$1,200.00 Delete
Housing Sep 8, 2026 Expense

Bus pass −$45.00 Delete
Transport Sep 6, 2026 Expense

Weekly groceries −$92.40 Delete
Food Sep 4, 2026 Expense

Monthly salary +$3,200.00 Delete
Salary Sep 1, 2026 Income

Monthly summary

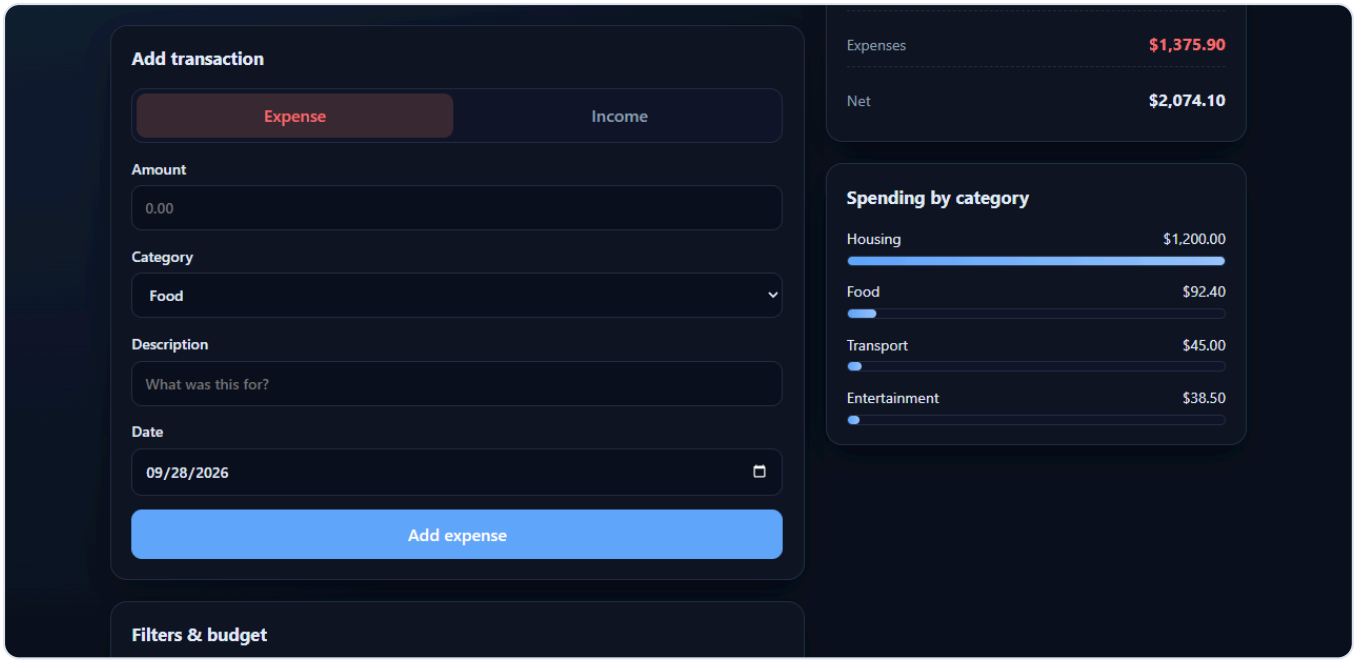
September 2026

Income	$3,450.00
Expenses	$1,375.90
Net	$2,074.10

Spending by category

Housing	$1,200.00
Food	$92.40
Transport	$45.00
Entertainment	$38.50

Transaction list and filters



Mobile layout

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Transactions (6)

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Freelance Sep 15, 2026

Income

Streaming **−$38.50** Delete

subscription

Entertainment Sep 12, 2026

Expense

Rent share **−$1,200.00** Delete

Housing Sep 8, 2026 Expense

Bus pass **−$45.00** Delete

Transport Sep 6, 2026

Expense

Weekly groceries **−$92.40** Delete

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Monthly salary **+$3,200.00** Delete

Salary Sep 1, 2026 Income

Monthly summary

September 2026

Income	**$3,450.00**
Expenses	**$1,375.90**
Net	**$2,074.10**

Spending by category

Housing	$1,200.00
Food	$92.40
Transport	$45.00
Entertainment	$38.50

9. Known limitations

- Currency display uses **USD** formatting only (no multi-currency)

- Transactions cannot be **edited** in place (delete and add again)
- Chart shows **expenses only**, for the selected month
- Data is stored only in this browser (localStorage)

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10. References

- React documentation: https://react.dev/
- Vite documentation: https://vite.dev/
- MDN localStorage: https://developer.mozilla.org/en-US/docs/Web/API/Window/localStorage

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This document supports the React course project submission alongside the GitHub repository and README.md.